

NCERT CURRICULUM FRAMEWORK

National Curriculum Framework — Teaching Guide for Primary Schools

Grades 1 to 8 — Core Subjects Overview

About NCERT: The National Council of Educational Research and Training

(NCERT) sets the curriculum standards for schools across India.

This guide summarizes the key learning outcomes per grade and subject to help teachers plan aligned lessons.

MATHEMATICS — GRADE-WISE LEARNING OUTCOMES

GRADE 1

- Count and write numbers up to 100
- Add and subtract single-digit numbers
- Identify basic shapes: circle, square, triangle, rectangle
- Measure length using non-standard units

GRADE 2

- Count and write numbers up to 1000
- Add and subtract two-digit numbers with carrying
- Introduction to multiplication as repeated addition
- Identify and draw 2D shapes

GRADE 3

- Multiply single-digit numbers (tables up to 10)
- Divide using simple division facts
- Introduction to fractions: half, quarter, one-third
- Read and write time on a clock

GRADE 4

- Multiply and divide 2-digit numbers
- Fractions: compare and order simple fractions
- Introduction to decimals
- Perimeter of simple shapes

GRADE 5

- Fractions: add and subtract fractions with same denominator
- Decimals: place value up to hundredths
- Area and perimeter of rectangles and squares
- Introduction to angles: acute, obtuse, right angle
- Data handling: bar graphs and pictographs

GRADE 6

- Integers: positive and negative numbers
- Ratio and proportion
- Basic algebra: simple equations
- Geometry: triangles and their properties
- Introduction to percentage

GRADE 7

- Rational numbers
- Algebraic expressions
- Congruence of triangles
- Data handling: mean, median, mode
- Profit and loss, simple interest

GRADE 8

- Squares and square roots
- Linear equations in one variable
- Quadrilaterals and polygons
- Introduction to graphs
- Factorisation

SCIENCE — GRADE-WISE LEARNING OUTCOMES

GRADE 4

- Parts of a plant and their functions
- Animals and their habitats
- Food and nutrition basics
- Water: sources and uses

GRADE 5

- Photosynthesis: how plants make food
- The human body: bones and muscles
- Simple machines: lever, pulley, wheel
- Air and water pollution

GRADE 6

- The cell: basic unit of life
- Separation of substances
- Changes around us: physical and chemical
- The living world: characteristics of living things

GRADE 7

- Nutrition in plants and animals
- Weather and climate
- Acids, bases and salts
- Motion and time

GRADE 8

- Crop production and management
- Microorganisms
- Force and pressure
- Light: reflection and refraction

ENGLISH — GRADE-WISE LEARNING OUTCOMES

GRADE 3–4

- Read simple sentences with comprehension
- Write 3–5 sentences on familiar topics
- Learn 200+ new vocabulary words per year
- Basic grammar: nouns, verbs, adjectives

GRADE 5–6

- Read short paragraphs and answer questions
- Write a paragraph of 5–8 sentences
- Grammar: tenses, prepositions, conjunctions
- Comprehension: main idea and supporting details

GRADE 7–8

- Read unseen passages and summarize
- Write essays of 150–200 words

- Grammar: active/passive voice, reported speech
- Letter writing: formal and informal

SAMPLE LESSON PLAN — NCERT ALIGNED

Grade 5 Mathematics: Fractions

Learning Outcome (NCERT Grade 5 Maths):

Students will add fractions with the same denominator.

Lesson Objective:

By end of lesson, students can solve 5 fraction addition problems.

Introduction (5 min):

Draw a circle on board. Divide into 4 equal parts.

Ask: "If I colour 1 part, what fraction is coloured?" ($\frac{1}{4}$)

Ask: "If I colour 1 more part, how many are coloured now?" ($\frac{2}{4}$)

Main Teaching (20 min):

Write on board: $\frac{1}{4} + \frac{1}{4} = \frac{2}{4}$

Rule: When denominators are same, add only the numerators.

Do 3 examples on board with student participation.

Practice (15 min):

Students solve: $\frac{2}{5} + \frac{1}{5}$, $\frac{3}{8} + \frac{2}{8}$, $\frac{1}{6} + \frac{4}{6}$

Assessment (5 min):

Exit ticket: Solve $\frac{1}{3} + \frac{1}{3} = \underline{\hspace{1cm}}$

Source: National Curriculum Framework, NCERT, New Delhi

Ministry of Education, Government of India